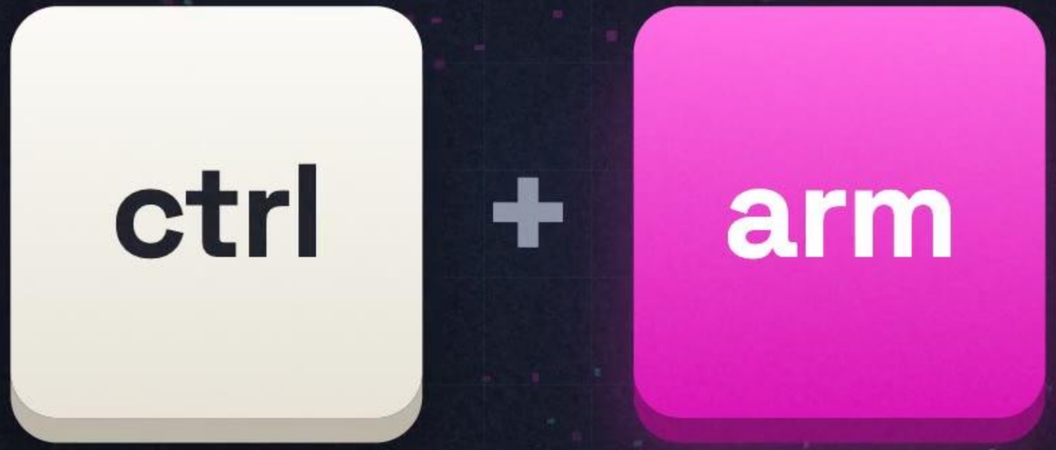


Find anything. For real.

Trova qualsiasi cosa, davvero. Parla italiano.

Grok vision + voice

THE PROBLEM

After every batch, someone checks the whole line by eye.

One leftover

A carton or label from the last batch ends up in the next pack: a mix-up, then a recall.

Slow, unproven

Checks by eye take time and leave no evidence of what was actually seen.

AI may only assist

Draft EU GMP Annex 22: a model can help, a person decides and signs.

Sources: European Pharmaceutical Review on Annex 22 · line-clearance vendor case studies

LIVE DEMO · 80 SECONDS

Three things to watch.

1

Find it

“Dov’è la lattina Red Bull?” Grok boxes it in the arm camera. The arm aims, then approaches; the operator confirms each stage.

2

What changed

We add an object. Trova says NOT CLEAR, names it and says why. The operator signs a PDF record.

3

Stop

Say “stop”. The arm holds at once. No model sits between the word and the brake.

Scoreboard live at /jury: objects found out of 10 · median time to find · false alarms

HOW IT WORKS

Grok decides what you mean. Code decides where the arm goes.

Pi camera
on the xArm 5, 1640×1232

Numbered boxes
plain OpenCV, no training

Grok vision
picks, asks back, explains

Calibration
hand–eye, pixel → robot mm

Motion gate
limits, speed, IK, STOP → arm



Real frame: plain code numbers every object


Approach view: the arm camera above the target

BUILT IN 36 HOURS

What runs today.

50+

commits from five people, all on GitHub

260+

automated tests, run before every commit

1.4 mm

median calibration error on replayed station data

3D

12-view orbit → mesh of a mouse, 104 × 67 × 39 mm

84

objects mapped in one 360° scan on the real arm — 12 photos, about a minute

EN · IT

ask by voice, by text or with a phone photo

Every arm move passes one safety gate; real motion needs an operator unlock and a confirmation per stage.

MARKET

A real market. A ready channel.

$15.8B

machine vision market in 2025, heading to $23.6B by 2030

$1.1B

pharma inspection machines, growing 7.5–10% a year

7,000+

SEA Vision systems already installed, €37.4M revenue in 2025

Business model: **software per line**, on the cameras and cobots a plant already owns. Same skill for warehouses, labs, hospitals and inspection.

Sources: MarketsandMarkets 2025 · Research and Markets 2025 · Marchesini Group / SEA Vision 2025

OUR ASK

A 30-day pilot on one SEA Vision packaging line.

Week 1

Mount the camera, calibrate, approve a reference.

Weeks 2–3

Run beside the operator: count catches and false alarms.

Week 4

Compare time and errors with manual clearance.

ctrl-arm

Find anything. For real.

Simone · Aman · Nawaz · Giacomo · Bhoumik

Thank you. Questions?